EconoCausal: 5-Day Plan to Complete Week 2

Project ITS/DSML/1147 | Infotact Solutions intern project | Plan window: Thu 15 Oct to Mon 19 Oct 2026

Before you start

Five working days from today (Thursday 15 Oct) is **15-19 Oct**, which finishes the day before the Mid Review window opens on 20 Oct. The Mid Review window is 20-27 Oct and covers Weeks 1 and 2.

The project image describes a different problem from the README:

	Project image (assigned brief)	README
Treatment	Price (continuous)	Binary discount
Outcome	Sales / units sold	Purchase
Goal	Optimal-price engine	Uplift and budget allocation

Day 1 settles which one you build. I recommend following the image, since it is the assigned brief. Confirm that choice with your mentor. If you stay with discount data, `discount_amount` can serve as a continuous treatment.

Day 1: Thu 15 Oct. Align data and set up

- Check that your dataset has price (or discount amount), units sold, cost price, competitor price, ad spend, inventory and date. If columns are missing, extend `generate_data.py` and keep seed 42. Store the true price effect so you can check whether DML recovers it.
- Re-run preprocessing and tests.
- Run `pip install econml` and add it to `requirements.txt`. If it fails on Python 3.14, make a 3.11 or 3.12 venv.
- Create each member's branch, submit the Work Distribution Document, and **make your first commit today**.

- Open the GitHub commit calendar and count how many distinct days you have already committed since 9 Oct. This tells you how far you are toward the 10 days needed for Mid Review.

Done when: `econml` imports, the dataset has treatment, outcome and confounders, and tests pass.

Day 2: Fri 16 Oct. Core DML estimator

Track	Tasks
Causal	In `src/causal/dml_estimator.py`: train/test split, then an outcome model E[Y|X] and a price model E[D|X] (Random Forest or XGBoost per the image). Cross-fit with 5 folds, compute the residuals, and fit `LinearDML`. Use log(units) against log(price) so the coefficient reads as price elasticity.
Evaluation	Record MAE, RMSE and R^2 for both nuisance models on the test set, and compare against a naive OLS of sales on price.
Dashboard	Build Plotly components for actual vs predicted sales and price vs demand.

Done when: you have an average price effect with a confidence interval and a naive-vs-DML comparison table.

Day 3: Sat 17 Oct. Heterogeneous effects and diagnostics

Track	Tasks
Causal	Add `CausalForestDML`, or `LinearDML` with a featurizer, to get effects by segment (season, product, region).

Track	Tasks
Diagnostics	In `diagnostics.py`, check that price residuals have real variation (the continuous version of overlap). Check residuals are not still correlated with X. Run DoWhy refutations: placebo treatment, random common cause, data-subset. Compare results across learners and fold counts.
Validation	If you stored the true effect, report how closely DML recovers it. Save all results to JSON.

Done when: `diagnostics.py` runs end to end and writes its results file.

Day 4: Sun 18 Oct. Dashboard, API and database

Track	Tasks
Dashboard	Fill the Causal page: DAG, effect distribution, price-demand curve with confidence band, segment effects, and filters.
API	Build FastAPI with `/health`, `/data/validate` and `/causal/estimate`.
Database	Store run metadata and results in SQLite. The Mid Review explicitly scores backend/API and database integration.

Done when: the dashboard shows real estimates, not placeholders, and the API returns them.

Day 5: Mon 19 Oct. Tests, report and merge

- Add tests for the estimator, diagnostics and API, then run the full `pytest` suite.
- Write `reports/week2_dml_report.md` with the causal question, assumptions, results with intervals, and limitations. Make no profit or ROI claims yet.
- Update the README with Week 2 run commands and an acceptance checklist.
- Team Lead merges all member branches into main and tags `week2`.

- Rehearse the viva. Each member should be able to explain residualization, why cross-fitting is used, and what the DAG assumes. Draft the social media video script.

Done when: main is green, the report is in the repo, and the demo runs from a clean clone.

Rules for all 5 days

- Every member commits meaningfully to their own branch each day. Empty, single-character or duplicate commits do not count.
- The Final Review needs a commit on every one of the 20 days before it, so keep this streak going past 19 Oct without a gap.
- Week 3 (the Phase 4 pricing engine, which finds the optimal price from predicted sales, revenue and profit) starts on 20 Oct, so do not pull it into this week.